

Probability & Statistics – The **Backbone** of **AI/ML**

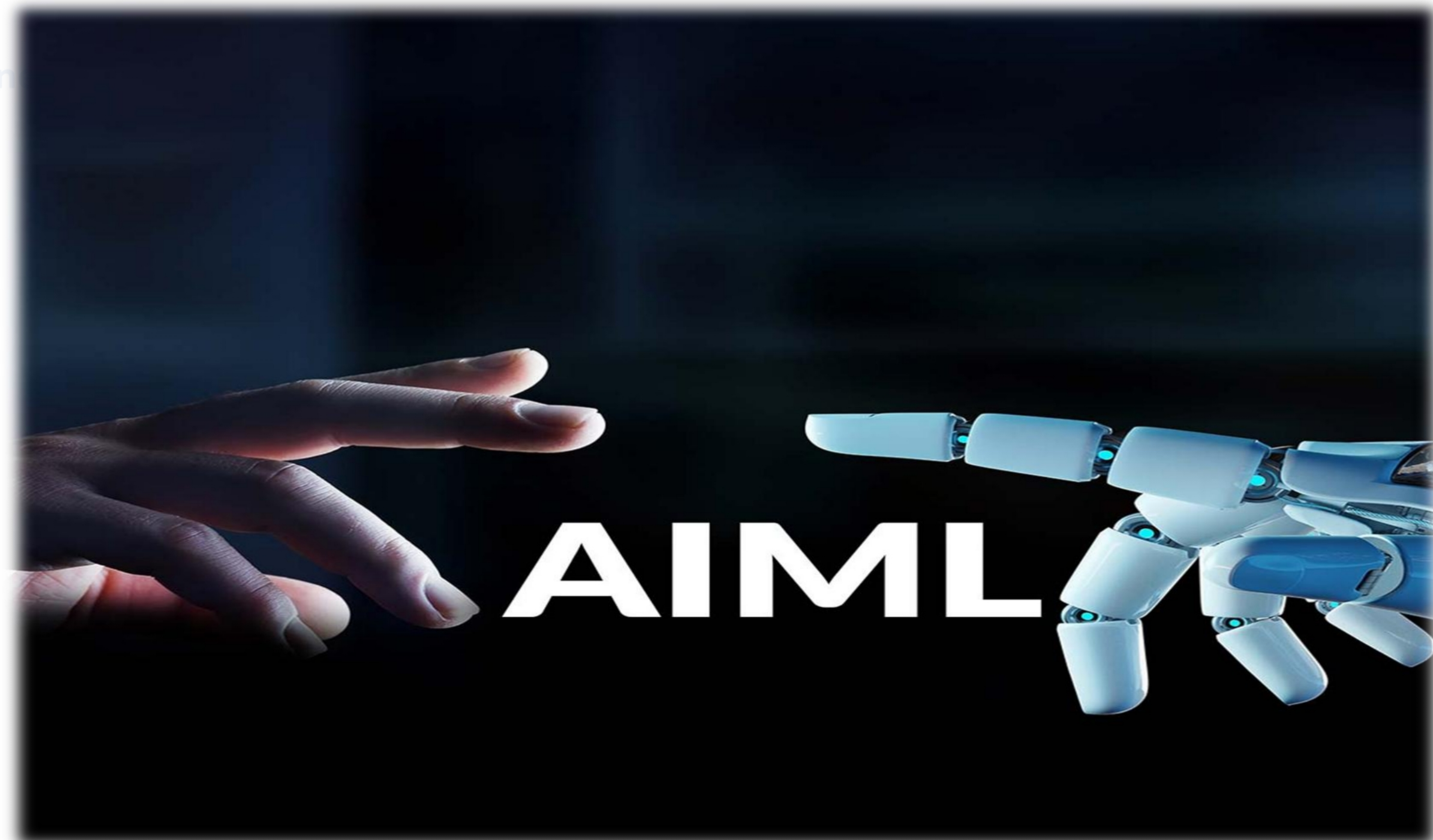
Linear Algebra – The Backbone

TOPIC :

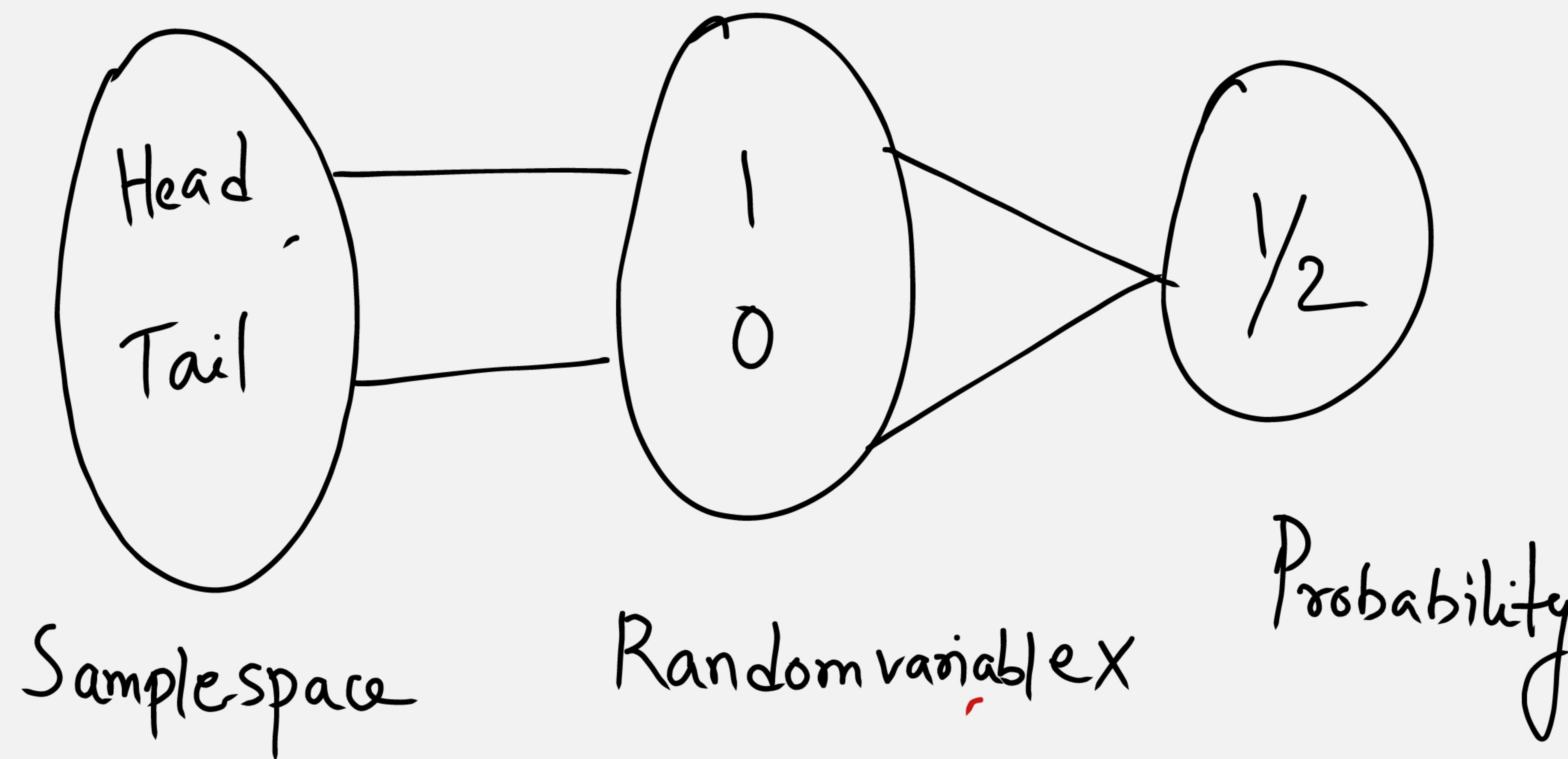
Random variable & Distribution -

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Random variable
(Coin Toss)



Random variable as a
function that maps
from the Sample space
of an experiment to the
real numbers,
Mathematically,
 $X: S \rightarrow \mathbb{R}$

Random variable

Continuous
Random
variable

[Infinite
values]

PDF (prob. density
Func)

$$* 0 \leq f(x) \leq 1$$

$$\int f(x) dx = 1$$

over all
values of x

Discrete
Random
variable

[takes on a finite
values]

PMF (Probability Mass Func)

$$* 0 \leq p \leq 1$$

$$* \sum p(x_i) = 1$$

Problem :-

Find the value of $P(1 < X < 2)$.

$$f(x) = Kx^3; 0 \leq x \leq 3$$

otherwise $f(x)$ is density function.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int f(x) dx = 1$$

$$\int Kx^3 dx = 1$$

$$\Rightarrow K \int x^3 dx = 1 \Rightarrow K \left[\frac{x^4}{4} \right] = 1$$

Given Interval, $0 \leq x \leq 3 = 0$

$$K [3^4 - 0^4] / 4 = 1$$

$$K \left(\frac{81}{4} \right) = 1$$

$$K = \frac{4}{81}$$

Thus,

$$P(1 < X < 2) = K \left[\frac{x^4}{4} \right]$$

$$P = \frac{4}{81} \times \frac{16 - 1}{4} = \frac{15}{81}$$



Problem :-

Find the mean value for the continuous random variable, $f(x) = x^2$, $1 \leq x \leq 3$

Given, $f(x) = x^2$, $1 \leq x \leq 3$

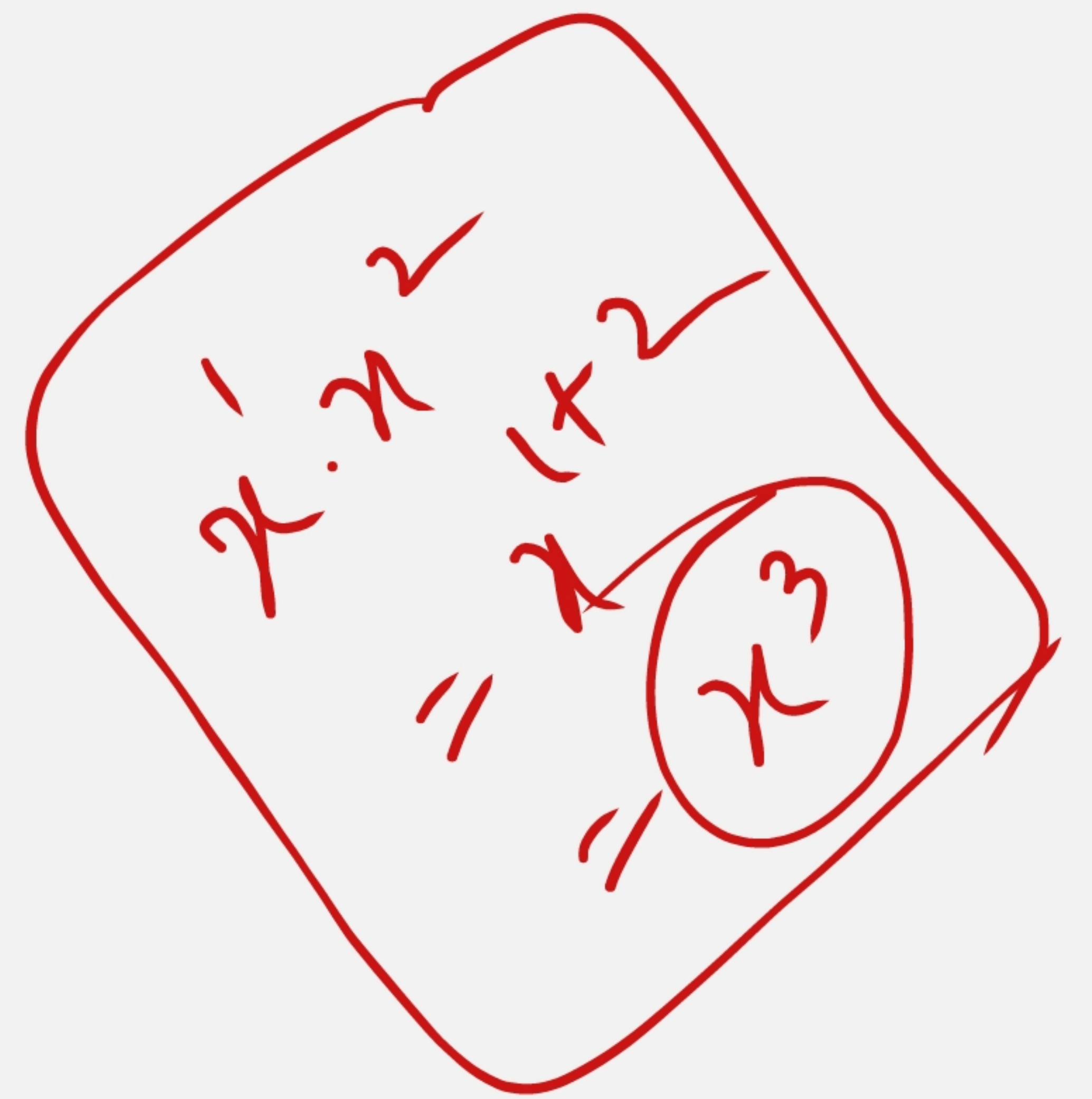
$$E(x) = \int_1^3 x \cdot f(x) dx$$

$$= \int_1^3 x \cdot x^2 dx$$
$$= \int_1^3 x^3 dx$$

$$= \left[\frac{x^4}{4} \right]_1^3$$

$$= \frac{1}{4} \times (3^4 - 1^4)$$

$$= \frac{1}{4} \times (81 - 1) = \frac{1}{4} \times 80 = 20$$



Problem :-

Find the mean value for the continuous random variable, $f(x) = e^x$, $1 \leq x \leq 3$

$$f(x) = e^x, 1 \leq x \leq 3$$

$$E(x) = \int_1^3 x \cdot e^x dx$$

$$= \left[x \cdot e^x - e^x \right]_1^3$$

$$= \left[e^x (x-1) \right]_1^3 = e^3(2) - e(0) \text{ Ans-}$$

Problem

Given the discrete random variable X with the following prob. distribution:

X	1	2	3	4
$P(X)$	0.1	0.2	0.4	0.3

Find the mean value (or expected value) of the random variable X .

$$E(X) = \sum x \cdot P(x)$$

$$= 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.4 + 4 \times 0.3$$

$$= 2.900 \checkmark$$

Problem :-

X is a discrete random variable having the prob. mass funⁿ:

$x:$	0	1	2	3	4	5	6
$p(X=x):$	0	K	$2K$	$3K$	$4K$	$5K$	$6K$

① Since the above dis. is probⁿ so,

$$0 + K + 2K + 3K + 4K + 5K + 6K = 1$$

$$K = \frac{1}{21}$$

① Determine the constant K .

② Find $p(X < 5)$ and $p(X \geq 4)$

$$\textcircled{II} P(X < 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$\textcircled{\text{upto } -4} = 0 + K + 2K + 3K + 4K = 10K = 10 \times \frac{1}{21} = \left(\frac{10}{21}\right) \text{ A.}$$

$$\textcircled{III} P(X \geq 4) = P(X=4) + P(X=5) + P(X=6)$$

$$\textcircled{\text{start from } -4} = 4K + 5K + 6K = 15 \times \frac{1}{21} = \left(\frac{5}{7}\right) \text{ A.}$$

Problem:-

A random variable X has pdf function

$$f(x) = \begin{cases} \frac{1}{4} & , -2 < x < 2 \\ 0 & , \text{elsewhere} \end{cases}$$

Solⁿ:-

$$\textcircled{i} P(X < 1) = \int_{-\infty}^1 f(x) dx$$

Find - $\textcircled{i} P(X < 1)$ ✓

$\textcircled{ii} P(|X| > 1)$ ✓

$\textcircled{iii} P(2X + 5 > 7)$

$[-1, 1]$

$$= \int_{-2}^1 \frac{1}{4} dx$$

$$= \frac{1}{4} [x]_{-2}^1 = \frac{1}{4} (1 + 2) = \frac{3}{4} \checkmark$$

Formula

$$\int_{-\infty}^{\infty} f(x) dx$$

Real line

$$\textcircled{\text{ii}} \quad P(|X| > 1) = P(X < -1) + P(X > 1)$$

$$= \int_{-2}^{-1} f(x) dx + \int_1^2 f(x) dx$$

$|x| > 1$

$$= \int_{-2}^{-1} \frac{1}{4} dx + \int_1^2 \frac{1}{4} dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\textcircled{\text{iii}} \quad P(2X+5 > 7) = P(X > 1) = \int_1^2 \frac{1}{4} dx = \frac{1}{4}$$

Distribution

Continuous

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Discrete
Distribution

$$\sum p(x) = 1$$

$$E(x) = \sum x \cdot p(x)$$

$$V(x) = E(x^2) - [E(x)]^2$$
$$= \sum x^2 p(x) - \left(\sum x p(x) \right)^2$$

Types of Distribution :-

Discrete Distribution

- (I) General Discrete Distribution
 - (II) Binomial Distribution
 - (III) Hypergeometric distribution
 - (IV) Geometric Distribution
 - (V) Poisson distribution
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